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Renouf, Marguerite A., Kenway, Steven J., Lam, Ka Leung, Weber, Tony, Roux, Estelle, Serrao-Neumann, Silvia, Choy, Darryl Low, & Morgan, Edward A.
(2018)
Understanding urban water performance at the city-region scale using an urban water metabolism evaluation framework.
Water Research, *137*, pp. 395-406.

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https://doi.org/10.1016/j.watres.2018.01.070

Elsevier Editorial System(tm) for Water Research

Manuscript Draft

Manuscript Number: WR40342R1

Title: Understanding urban water performance at the city-region scale using an urban water metabolism evaluation framework

Article Type: Research Paper

Keywords: urban metabolism; water mass balance; water efficiency; wastewater recycling; stormwater; water sensitive cities

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Abstract: Water sensitive interventions are being promoted to reduce the adverse impacts of urban development on natural water cycles. However it is currently difficult to know the best strategy for their implementation because current and desired urban water performance is not well quantified, particularly at the city-region scale at which strategic urban planning occurs. This work aimed to fill this gap by quantifying the water performance of urban areas within the city-regions using 'urban water metabolism' evaluation, to inform decisions about water sensitive interventions. To do this we adapted an existing evaluation framework with new methods. In particular, we developed and applied rigorous system boundary definitions based on consistent land use data, and for estimating natural hydrological flows by combining the land use data with hydrological flow partitioning (HFP) factors. The water performance criteria were water efficiency of the urban systems (in terms of water extracted externally) and hydrological performance of the urban systems (the degree natural hydrological flows have changed relative to pre-urbanised state). We compared these performance criteria for urban systems within three Australian city-regions (South East Queensland, Melbourne and Perth metropolitan areas) under current conditions, and after implementation of example water sensitive interventions (demand management, rainwater/stormwater harvesting, wastewater recycling and increasing perviousness). The respective water efficiencies in terms of water sourced externally were found to be 79, 90 and 133 kL/capita/yr. In relation to hydrological performance, stormwater runoff relative to pre-urbanised flows was of most note, estimated to be 2-, 6- and 3- fold, respectively. The estimated performance benefits from water sensitive interventions suggested different priorities for the urban systems within each of city-regions, and that combined implementation of a range of interventions may be necessary to make substantive gains in water metabolism performance. We concluded that the framework is suited to initial screening of the type and scale of water sensitive interventions needed to achieve desired water performance objectives.

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Understanding urban water performance at the city-region scale using an urban water metabolism evaluation framework

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KEYWORDS

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ABSTRACT

Water sensitive interventions are being promoted to reduce the adverse impacts of urban development on natural water cycles. However it is currently difficult to know the best strategy for their implementation because current and desired urban water performance is not well quantified. This is particularly at the city-region scale at which strategic urban planning occurs. This work aimed to fill this gap by quantifying the water performance of urban areas within the city-regions using 'urban water metabolism' evaluation, to inform decisions about water sensitive interventions. To do this we adapted an existing evaluation framework with new methods. In particular, we developed and applied rigorous system boundary definitions based on consistent land use data, and for estimating natural hydrological flows by combining

the land use data with hydrological flow partitioning (HFP) factors. The water performance criteria were water efficiency of the urban systems (in terms of water extracted externally) and hydrological performance of the urban systems (the degree natural hydrological flows have changed relative to pre-urbanised state). We compared these performance criteria for urban systems within three Australian city-regions (South East Queensland, Melbourne and Perth metropolitan areas), under current conditions, and after implementation of example water sensitive interventions (demand management, rainwater/stormwater harvesting, wastewater recycling and increasing perviousness). The respective water efficiencies in terms of water sourced externally were found to be 79, 90 and 133 kL/capita/yr. In relation to hydrological performance, stormwater runoff relative to pre-urbanised flows was of most note, estimated to be 2-, 6- and 3- fold, respectively. The estimated performance benefits from water sensitive interventions suggested different priorities for each region, and that combined implementation of a range of interventions may be necessary to make substantive gains in water metabolism performance. We concluded that the framework is suited to initial screening of the type and scale of water sensitive interventions needed to achieve desired water performance objectives.

1. INTRODUCTION

Innovations in total water cycle management and water-sensitive urban design are being promoted to secure water supplies and to reduce the impacts of urban development on the natural water cycle (ADB 2016, OECD 2015, UK Water Partnership 2015), and articulated in concepts such ‘water-sensitive cities’ (Wong and Brown 2009) and ‘water wise cities’ (IWA 2016). These include water resource management interventions such as improved water use efficiency and diversification of water supplies (harvesting of rainwater and stormwater runoff, wastewater recycling) (Campisano et al. 2017, Khastagir and Jayasuriya 2010, Sharma et al. 2013), and planning interventions such as water sensitive urban designs (WSUD) and

56 greater green space (Coutts et al. 2013, Feng et al. 2016, Sharma et al. 2016). We collectively
57 refer to these as water-sensitive interventions.
58 To understand how best to employ these interventions, water managers and urban planners
59 need quantification of urban water performance (i.e., how efficiency water is being utilised,
60 how the natural water cycle is altered) (Chesterfield et al. 2016). They need to know what the
61 current urban water performance is, what the desired water performance should be, and the
62 extent to which various interventions can influence performance. Quantification of urban
63 water performance is an evolving science with a range of approaches described in the
64 academic literature (Behzadian and Kapelan 2015a, b, Makropoulos et al. 2008, Renouf and
65 Kenway 2016, Rozos and Makropoulos 2013, Urich et al. 2013, van Leeuwen et al. 2012,
66 Venkatesh et al. 2014).
67 One approach that has been shown to be useful is 'urban water metabolism' evaluation, in
68 particular for providing a big-picture perspective of urban water performance. It is based on
69 the concept of urban metabolism (Wolman 1965), which has been operationalised to evaluate
70 the material and energy flows through urban areas (examples are Kennedy et al (2007) and
71 Newman (1999)), and specifically adapted for evaluating urban water performance (Kenway
72 et al. 2011, Renouf and Kenway 2016). It generates a comprehensive account of all flows of
73 water (natural and anthropogenic) between an urban area and the supporting environment to
74 produce an urban water mass balance, from which indicators of 'water metabolism'
75 performance are derived. These indicators relate to metabolic aspects such as the overall
76 water efficiency, the degree of supply internalisation, and the extent to which natural
77 hydrological flow are altered (Renouf et al. 2017).
78 The approach has been used to characterise the water balances and water performances of
79 cities and metropolitan areas (Bhaskar and Welty 2012, Kenway et al. 2011, Marteleira et al.
80 2014, Thériault and Laroche 2009) and to compare alternative urban water servicing options

(Farooqui et al. 2016). In this past work, the system boundary and spatial scale of analysis has been based on individual cities or urban areas, with the system boundary generally dictated by data available for estimating water flows (Kennedy et al. 2015). Urban system boundaries have generally not been systematically defined, and importantly for this research, not aligned to the urban planning context. We propose that a closer alignment to urban planning and water management processes will allow the knowledge generated to better inform decision making (Serrao-Neumann et al. 2017).

In the context of Australia, the 'city-region' is the spatial scale at which strategic urban planning often occurs, especially for the fast-growing capital city-regions (Low Choy and Minnery 1994). City-regions are larger in geographic scale than individual cities, often composed of multiple metropolitan areas, regional centres, and hydrological catchments, and including not only urban areas but also surrounding the peri-urban and rural areas. It is proposed that urban water resources should be managed at this scale (Plummer et al. 2011, Vietz et al. 2016).

A wider geographic perspective gives water managers and urban planners a broader picture of the implications of water sensitive interventions. For example, a previous study that evaluated alternative water servicing option for a new urban development (Farooqui et al. 2016) observed that the benefits of utilising the recycled wastewater outside the urban area (in an adjoining rural area) could not be accounted for because the boundary was constrained to just the urban areas. A broader system boundary that captures the wider influence of water sensitive interventions will help optimise water performance for the city-region overall.

Therefore, the main aim of this work was to adapt the urban water metabolism evaluation approach for application at the city-region scale, and to test how application at this scale can inform decision about water-sensitive interventions. The specific question asked was: can

quantification of 'urban water metabolism' performance at the city-region scale inform decisions about water sensitive interventions?

The evaluation approach we adapted was based on an urban water mass balance method originally described by Kenway et al (2011), and further developed into an evaluation framework by Farooqui et al (2016). Adaption of this approach at this larger urban scale involved defining the urban systems within the city-region, and then estimating the exchanges of water between the urban systems and the wider supporting city-region. In the absence of an established protocol for defining urban system boundaries for metabolism studies (Alberti et al., 2017), it was necessary to develop a new method for doing this using land use data. It was also necessary to develop new methods for estimating water flows based on the land use data, so they align to the defined boundary. This meant we fitted the data to the defined boundary, rather than fitting the boundary to the data, which has been the practice in the past. Therefore a contribution of this work has been to advance the existing evaluation framework by adding capacity for defining the urban system boundary in a systematic and repeatable way.

To answer our research question the adapted evaluation framework was used to compare the water metabolism performance of three Australian case study city-regions, under current conditions, after implementation of a range of water sensitive intervention, and also under pre-urbanised conditions (as a point of reference). This allowed us to test how the information could inform decisions about urban water management.

The gaps this research fills, and the novel contributions are i) the application of urban water metabolism evaluation at the city-region scale; ii) a systematic method for defining urban system boundaries allowing consistent comparisons, and iii) methods for estimating urban water flows to align with a defined urban system boundary.

The work substantially develops urban water metabolism evaluation by providing more rigorous methods to enable consistent comparisons of the water performance of urban areas. It

demonstrates how the refined methods can generate water performance information about (i) pre-existing (ii) current and (iii) future urban conditions, at a scale that aligns with strategic urban and water planning processes. It shows for the first time how this approach can enable a better understanding of the roles of water sensitive interventions in achieving water performance objectives for city-regions, including the establishment of urban water performance goals, to help manage increasingly water-stressed city-regions.

2. MATERIAL AND METHODS

2.1 Method overview

The urban water metabolism evaluation framework used here is an adaptation of a framework developed by Farooqui et al (2016) to generate performance indicators proposed by Renouf et al (Renouf et al. 2017). The adapted framework has four method components: i) definition of the urban system boundary using spatial land use data, ii) estimation of water flows, iii) compilation of the flow data into a water mass balance, and iv) derivation of indicators of urban water metabolism performance from the water mass balance data (Figure 1). The first is a new component developed by this research to systematically define the urban system. For the second component, the methods for estimating water flows were refined in this research so they align to the defined urban system boundary. The framework was applied to three Australian case study city-regions to characterise and compare their water metabolism performance under current conditions and after implementation of a range of water sensitive interventions. In the sections that follow, we describe each of the steps described in the framework (Figure 1).

Insert Figure 1 near here

The three case study Australian city-regions were South East Queensland (SEQ), the Melbourne metropolitan region (MEL) and the Perth metropolitan region (PER) (Figure 2), which are a reasonable representation of the different climatic, typological, and urban form characteristics present in urban Australia (Table 1). They are examples of increasingly water-stressed city-regions, which face the challenges of planning for future population growth with reduced and more erratic water supply due to climate change (Gooda and Voogt 2012, Risbey 2011, Schandl et al. 2008). The sections that follow describe each of the steps of the framework.

Insert Figure 2 near here

Insert Table 1 near here

2.2 System boundary

2.2.1 Spatial boundary of the urban systems within the city-regions

The spatial extent of the case study city-regions (Figure 2) were derived from urban planning documents, more specifically the strategic regional plans for growth management (Queensland Government 2016, Victorian Government 2014, Western Australian Government 2015). They are shown in Figure 2 and in more detail in the Supplementary Material (Figures S1, S2 and S3). Urban water supplies are sourced from the hydrological catchments within these regions, so the city-regions also represent the supporting environments from which water supplies are drawn.

The 'urban systems' within the city-regions were then defined. We sought to define the urban systems in a systematic and repeatable way and to align the data to that definition.

178 Using population density to define urban development, as used by the OECD (2012), was
179 considered. However, because population data can have relatively coarse resolution (being
180 linked to administrative areas), it is difficult to reliably define the spatial extent of urban
181 development (Maheshwari and Thoradeniya 2016).

182 The approach we arrived at was to use land use spatial data to categorise land uses, from
183 which we identified area that are urban in the nature. Spatial data for land uses within city-
184 regions are available at very high level of resolution ($100m^2$) in geographical information
185 systems (GIS). This allowed detailed spatial statistical analysis using ArcGIS 10.3 software.
186 The GIS land use data for the case study regions was sourced from the Catchment Scale Land
187 Use of Australia (CLUM) database for the year 2014 (ABARE 2015), and categorised using
188 the Australian Land Use and Management Classifications (ALUM) system (ABARE 2013).
189 Furthermore, the land use information can be linked to imperviousness for estimating natural
190 hydrological flows in subsequent steps (see Section 2.3.2).

191 A limitation of using spatial land use data is that the distribution of land uses at the
192 periphery of urban areas is very fragmented, making it difficult to define an actual boundary
193 'envelope' in a consistent and reproducible way. To address this, the 'urban footprints'
194 defined in strategic regional plans were also referred to. 'Urban footprints' define the agreed
195 extent of urban development for a strategic planning horizon (around 25 years). It provides an
196 agreed boundary for what is regarded by the planning sector as urban (and future urban), and
197 a useful 'envelope' for analysing land use using GIS.

198 We identified land uses within the urban footprint to identify the areas that can be
199 considered part of the urban system. The land use categories considered to be urban in nature
200 are detailed in the Supplementary Material, but broadly included residential, commercial,
201 industrial, recreational, cultural uses and natural areas that are part of the urban landscape.

202 Peri-urban areas (those categorised as rural residential) were also included as part of the
203 'urban system'. They are the lower-density fringe on the periphery of urban areas, and are
204 present both within and outside the urban footprtint perimeter. They were included because
205 they are a significant feature of Australian city-regions, and many water sensitive
206 interventions (such as stormwater harvesting and wastewater recycling) will occur in or
207 influence these areas due to open space available.

208 The resulting urban system areas identified for each of the city-regions are shown in Figure
209 2, and in more detail in the Supplementary Material (Figures S1, S2 and S3). They represent
210 the system boundaries of the analyses. The calculated land areas for the city-regions and
211 urban systems within them are summarised in Table 1. Vertically, the urban system extends
212 from rooftop and tree tops, to the root zones of trees (assumed to be 1 meter below ground)
213 (Kenway et al. 2011).

214 *2.2.2 Technical boundary to distinguish between water extracted from 'external' versus*
215 *'internal' sources*

216 Our interpretation of urban water metabolism suggests a desire to reduce reliance on
217 freshwater extracted externally from the environment, and the indicator of water efficiency
218 generated by the framework is in terms of water extracted externally (discussed in Section
219 2.5). Application of the framework at the larger city-region scale highlighted the need to
220 distinguish between water sourced the environment ('external' water) versus water sourced
221 from within the urban system ('internal' water) (Renouf et al. 2017). This differentiation is in
222 terms of the technical system rather than spatial location, and is analogous with technical
223 system boundary definition well-established in environmental life cycle assessment for
224 production systems (Rebitzer et al. 2004).

225 Centralised water supplies sourced from surface or ground waters are regarded as
226 'external' water sources, because they originate from environment stocks. Desalinated water

was also categorised as ‘external’ water, because the water originates from seawater, even though the freshwater is generated within the urban system. In comparison, decentralised supplies harvested from within the urban system, are regarded as ‘internal’ water sources. The terms ‘centralised’ and ‘decentralised’ do not adequately describe this distinction, as they refer to the way water supplies are managed rather than where they come from. However, we have retained the use of these terms to be consistent with industry terminology, and clarify our meaning by calling them centralised (external) water versus decentralised (internal) water.

Groundwater would generally be considered an ‘external’ water source. In the PER case, groundwater from the region’s very high water tables is an important urban water supply, and infrastructure is being built to use aquifers as water storage ‘reservoirs’ with managed recharge, including injection of highly-treated wastewater (Water Corporation n.d.). Aquifers that function as such could be regarded as ‘internal’ water sources. However, we categorised water from Perth’s groundwater aquifers as ‘external’ water for consistency.

2.3 Estimation of water flows

The water flows estimated for the analyses represent exchanges of water between the urban system areas and the supporting city-region, as annually-averaged flows (GL/yr). The categorisation and nomenclature of the individual flows (see Figure 1) was based on the original evaluation framework of Farooqui et al (2016), but with the further refinements of Renouf et al (2017) to differentiate between water sourced internally and externally (see Section 2.2.2). In order for the water flow data to align with the defined urban system boundaries at the city-region scale, different methods were developed and data sources were used to those used in the Farooqui et al (2016) framework.

2.3.1 Estimation of anthropogenic flows

Anthropogenic flows are those managed by urban water infrastructure (orange arrows in Figure 1), i.e., centralised (C) and decentralised (D) water supplies, and wastewater that is

252 discharged (WW) or recycled (Re). Data for the centralised anthropogenic flows were mostly
253 sourced from the Australian Bureau of Meteorology's Urban Water Accounts (BOM UWA)
254 database (BOM 2016), which generates Australia's National Water Accounts (BOM 2015). It
255 compiles data on annual volumes of water supplied by bulk water providers and water
256 retailers (potable and non-potable to residential, commercial and industrial), and wastewater
257 generated, treated and recycled by wastewater treatment utilities. The data derived from the
258 database for the year of the study (2013/2014) was aligned to the defined urban system
259 boundaries by only including data for the water utilities that service these areas.

260 The BOM UWA database does not report decentralised supplies from harvested rainwater,
261 stormwater or bore water. These were instead derived from published reports, or estimated
262 from surveys (Table 2). The amount of harvested rainwater was estimated from the amount of
263 precipitation, average household roof area, number of households, tank volumetric reliability
264 (Umapathi et al. 2012), and tank penetration (ABS 2013). Data was not available for dam and
265 bore water collected or extracted in peri-urban areas not serviced with the potable urban water
266 supply, which was an omission from the water balance.

267 *Insert Table 2 near here*

268 *2.3.2 Estimation of natural hydrological flows*

269 Natural hydrological flows are those in the natural water cycle (blue arrows in Figure 1),
270 i.e., precipitation (P), stormwater runoff (SW), infiltration to groundwater (G), and
271 evapotranspiration (ET). Water that flows through the urban system boundary in natural
272 watercourses (creeks, rivers, aquifers etc.) are not accounted for as they are considered part of
273 the environment.

274 To estimate the size of P, SW, G and ET we first made assumptions about the fractions of P
275 that become SW, G, and ET. They are termed hydrological flow partitioning (HFP) factors.
276 Runoff and infiltration coefficients were derived for each city-region using the U.S. Soil
277 Conservation Service runoff curve number method (USDA Soil Conservation Service 1986)
278 and the Australian Bureau of Meteorology's (BOM) Australian Landscape Water Balance
279 database (BOM 2017), respectively . Runoff coefficients refer to the amount of SW relative to
280 P, i.e. the fraction of P that becomes SW. The runoff curve number method used to derive
281 them used information about soil type, land use type (imperviousness) and daily precipitation
282 for each city region. After deducting the estimated SW fraction, the remainder was assumed
283 to either evapotranspire (ET) or infiltrate (G). The amount that infiltrates (G) was estimated
284 using the average infiltration coefficient for each city region, derived from the amount of deep
285 drainage reported in the BOM landscape water balance. The remainder was then assumed to
286 evapotranspire (ET). The resulting HFP factors for each land use type are reported in Table 3,
287 and further details of their derivation are included in the Supplementary Material.

288 To estimate the total annual volumes of these flows (GL/yr), the HFP factors for each land
289 use type (Table 3) were multiplied by the areas of respective land use types present within the
290 urban system boundary (See Supplementary Material). This method aligned with the urban
291 system boundary because it used the same land use data used to define the system boundary.

292 *Insert Table 3 near here*

293

294 **2.4 Urban water mass balance**

295 The estimates of anthropogenic and natural flows (GL/yr) were brought together into
296 the urban water mass balance (equations 1 and 2 below), using the method as described in the
297 original framework of Farooqui et al (2016) (see Figure 1). The aim was to achieve a mass

balance, such that total inflows equal total outflows, plus any changes in storage, thereby ensuring a comprehensive and accurate account (equations 1 and 2). Changes in storage were assumed to be zero. In this context storage refers to soil moisture and water stored in reservoirs within the urban system. So no changes in storage assumes stable climatic conditions and no influence on this stored water.

$$Input\ (Qi) = Output\ (Qo) + \Delta S \quad (1)$$

$$(P + C + D + Re) = (ET + SW + WW + G + Re) + \Delta S \quad (2)$$

Where:

P = unharvested precipitation falling in the urban boundary, i.e. total precipitation less any rainwater or stormwater harvested within the urban system area

C = total centralised (external) water supplies, which includes surface waters (C_s), groundwater (C_g) and desalinated water (C_d)

D = total decentralised (internal) water supplies harvested from within the urban system area, which includes harvested rainwater (D_r) and harvested surface water runoff (D_s)

ET = evapotranspiration from the urban system areas

SW = run-off of surface water / stormwater discharged from the urban system areas (not including that which is harvested)

WW = wastewater discharged from the urban system areas (total wastewater generated less that which is recycled)

G = infiltration to groundwater

Re = reuse / recycling of wastewater

ΔS = change in the stored water volume within the defined urban system areas

2.5 Urban water metabolism indicators

321 The following urban water metabolism performance indicators were derived from the water
322 mass balance data using the equations in Table 4. They were based on an set of indicators
323 proposed by Renouf et al. (2017) and previously demonstrated by Farooqui et al. (2016).

324 **Urban water efficiency** is an indicator of the overall efficiency of water use for the
325 defined urban system, in terms of water sourced 'externally' from outside the urban system. It
326 is expressed as a rate of 'external' water withdrawal per inhabitant per year (kL/capita/yr). It
327 is an outcome of both the urban water demand, and the degree of supply internalisation
328 (rainwater and stormwater collection, or wastewater recycling).

329 **Hydrological performance** is an indicator of the degree to which natural hydrological
330 flows have increased or decreased relative to a defined pre-urbanised reference state. It is the
331 ratio of post-to pre-urbanised annual flows for stormwater runoff, evapotranspiration, and
332 infiltration to groundwater, and so is dimensionless. A ratio of greater than one means that the
333 magnitude of the annual flow is larger than pre-developed landscape, and a ratio of less than
334 one means it is smaller. The pre-urbanised state provides a useful point of reference against
335 which to relative performance and track change. Hydrological flows for the pre-urbanised
336 reference could be approximated by using the HFP factors for 'conservation and natural
337 environment' land use category.

338 *Insert Table 4 near here*

339 **2.6 Case studies of water sensitive interventions**

340 The 'current' case, against which water sensitive interventions were compared,
341 represents current conditions using data for 2013/14. The 'water sensitive' (WS) interventions
342 represent a selection of water sensitive initiatives being promoted in Australia, i.e., reducing
343 water demand, internal harvesting of water (rainwater, stormwater), recycling wastewater, and
344 increasing perviousness to reduce runoff and increase groundwater infiltration (Brown and

345 Farrelly 2009). The scale of the interventions were based on what might be reasonably
346 achievable in the context of the city-region. For example, in SEQ and MEL there is capacity
347 for harvesting rainwater / stormwater and recycling wastewater. However, for PER harvesting
348 of rainwater and stormwater is less economic feasible due to the limited rainfall and very high
349 perviousness due to sandy soils. In addition, PER has a high proportion of decentralised use
350 of groundwater in local bores, which currently function in a similar way to rainwater tanks for
351 outdoor water use. Therefore there is no internal harvesting scenario for PER. The mass
352 balances and water metabolism indicators for these interventions were generated by changing
353 the appropriate variables (Table 5).

354 – Reduced demand scenarios (WS1) were based on demand reductions that reduce the
355 centralised water supply in SEQ, MEL and PER by 8%, 18% and 15%, respectively,
356 taking into account demand reductions that have already occurred.
357 – Internal harvesting scenarios (WS-2) were based on a combination of rainwater harvesting
358 and stormwater harvesting in the SEQ and MEL cases only. Rainwater harvesting is based
359 on an increased percentage of houses with a rainwater tank installed, from 47% to 60% in
360 SEQ and from 31% to 40% in MEL. Stormwater harvesting is based on an increases
361 percentage of harvestable stormwater that is harvested from 0.04% to 5% in SEQ and
362 from 0.3% to 5% in MEL. The rainwater and stormwater harvesting combined results in a
363 reduction of centralised water supply by 12%, and 9%, respectively.
364 – Wastewater recycling scenarios (WS-3) were based on increasing the percentage of
365 wastewater that is recycled from 7% to 20% in SEQ, from 10% to 20% in MEL, and from
366 6% to 40% in PER. This results in a reduction of centralised water supply by 11%, 9%
367 and 10%, respectively.
368 – Increasing perviousness scenarios (WS-4) were based on increasing the perviousness in
369 the urban residential areas to be similar to that of rural residential areas.

370 'Pre-urbanised' conditions were also used as the point of reference for generating the
371 indicators of hydrological performance. and were assumed to be the natural landscape with
372 hydrological flows based on the HFP factors for the natural environment land use category
373 (see Table 3).
374 *Insert Table 5 near here*

375 **3. RESULTS**

376 The water mass balance results (Table 6) summarise our estimates of all water inflows
377 and outflows at the city-region scale, for the 'current' case (2013/14), for the water sensitive
378 (WS) interventions, and the pre-urbanised reference cases. It shows how the current scales of
379 flows have changed relative to the pre-urbanised reference state, and how they may change
380 with water sensitive interventions.

381 For SEQ and MEL, the majority of anthropogenic water inflow is potable water
382 sourced from surface waters, and supplied via the centralised distribution systems. They also
383 have only small amounts of decentralised water (from rainwater and stormwater, or from
384 wastewater recycling). For PER, the major anthropogenic inflow is centralised potable water
385 from groundwater and seawater desalination, but decentralised water from domestic bores is
386 also a substantial source. Compared to the pre-urbanised states, there has been an increase in
387 the total flow of water passing through the urbanised city-regions (Qi and Qo) of 10% for
388 SEQ, 20% for MEL and 49% for PER (Table 6). The noticeably higher change for PER can
389 be attributed to PER having a relatively smaller area compared to SEQ and MEL, which
390 distorts the ratio between rainfall and anthropogenic inflows.

391 **3.1 Water metabolism characteristics**

392 The results for the water metabolism indicators (Figures 3) provide a more meaningful
393 interpretation of the water mass balance results. The urban water efficiency results (Figures
394 3a-c) combine urban water demand (the total height of the bars), water supply internalisation
395 (the top portion), to generate an overall urban use of 'external' water (the bottom portion).
396 The hydrological performance indicators (Figures 4d-f) show how the current annual volumes
397 of stormwater runoff, evapotranspiration and infiltration compare to the expected "pre-
398 urbanised" hydrology.

399 *Insert Table 6 near here*

400 *Insert Figure 3 near here*

401 *3.1.1 South East Queensland (SEQ)*

402 SEQ currently has a relatively low water demand (92 kL/capita/yr) compared to the
403 other city-regions (Figure 3a). This is in part due to behaviour change embedded during a
404 preceding drought when severe water restrictions and successful water conservation
405 campaigns were imposed (Walton and Hume 2011). 14% of supply is currently sourced from
406 internal supplies (from wastewater recycling, and relatively small amounts of rainwater and
407 stormwater harvesting), making the amount sourced from the environment 79 kL/capita/yr.
408 Relative to pre-urbanised hydrological flows, surface water runoff was estimated to be 201%,
409 and groundwater infiltration was 92% and evapotranspiration was 91% (Figure 3b).

410 Of the water sensitive interventions analysed, internal harvesting of rainwater /
411 stormwater (WS2) and wastewater recycling (WS3) may have the greatest influence on the
412 extraction of 'external' water, with both reducing it from 79 to around 70 kL/capita/yr. In
413 comparison, residential demand reductions, which are quite modest due to SEQ's earlier
414 demand management successes, reduce overall environment water use to 72 kL/capita/yr. If

all of these interventions were combined, it would reduce to 54 kL/capita/yr. Increasing perviousness (WS4) to the degree analysed was predicted to reduce surface water runoff by 12%. The increase in groundwater infiltration and evapotranspiration from increased perviousness would be quite minor, increasing groundwater infiltration from 92% to 95%, and evapotranspiration from 91% to 94% of "pre-urbanised".

3.1.2 Melbourne (MEL)

Total water demand in MEL is 107 kL/capita/yr, and 16% is supplied is from internal water sources, making the amount of water sourced from the environment 90 kL/capita/yr (Figure 3c). In relation to hydrological flows, surface water runoff was estimated to be 584% of pre-urbanised flows, groundwater infiltration 95% and evapotranspiration 96% of the pre-urbanised state (Figure 3d). This high increase in runoff is due to MEL having high levels of imperviousness compared with the other city-regions.

Demand management (WS1) could lead to the greatest influence on the overall rate of 'external' water use, reducing it from 90 to 74 kL/capita/yr. In comparison, greater internalisation of water supply (WS2) and wastewater recycling (WS3) each reduces overall environment water use to 82 kL/capita/yr. If all of these interventions were combined, it would reduce to 58 kL/capita/yr. Increasing perviousness to the degree analysed was predicted to reduce surface water runoff from 584% to 472% of pre-urbanised flows. Relative groundwater infiltration would increase from 95% to 97%, and evapotranspiration from 96% to 98%.

3.1.3 Perth (PER)

PER has a much higher water demand than the other city-regions at around 182 kL/capita/yr. This has been attributed to access to alternative water sources in Perth

438 (desalination and household bores) which has reduced the extent of household water
439 conservation (Lam et al., 2016; Lindsay et al., 2017). 27% is supplied from internal water
440 sources, resulting in an overall rate of externally sourced water use of 133 kL/capita/yr
441 (Figure 3e). In this analysis, the relatively high level of desalinated water inputs was regarded
442 as an 'external' source.

443 In relation to hydrological flows, surface water runoff was estimated to be 336% of the
444 pre-urbanised state (Figure 3f). The hydrological impact of highly permeable soil in the
445 region was discussed by Barron *et al.* (2013). Their work shows a reduction in
446 evapotranspiration and an increase in runoff within the groundwater systems after
447 urbanisation. In our work, it is interesting to note that evapotranspiration was predicted to
448 increase relative to the pre-urbanised state, rather than decrease. This difference from
449 previous work is mainly because in our water balance, we considered anthropogenic water
450 flows that includes high outdoor water use for irrigation of lawns and gardens (Bhaskar et al.
451 2016).

452 The scale of demand management could reduce extraction of 'external' water, from
453 133 to 113 kL/capita/yr. Water recycling has a similar effect; reducing the 'external' water
454 demand to 119 kL/capita/yr. If both of these interventions were combined, it would reduce to
455 around 100 kL/capita/yr. Increasing perviousness to the degree analysed was predicted to
456 reduce surface water runoff from 336% to 125% of pre-urbanised flows. This is a
457 considerably higher reduction than observed in the SEQ and MEL cases because PER's
458 baseline runoff is smaller due to relatively high perviousness and sandy soil type. So the
459 relative improvement is amplified. Groundwater infiltration is expected to increase from 96%
460 to 99%, and evapotranspiration from 116% to 118%, relative to a pre-urbanised state.

4. DISCUSSION

The urban water metabolism evaluation of the three Australian city-regions provided a broad picture of current urban water performance, highlighting differences between the city-regions and suggesting different improvement priorities.

For example, the urban systems within the SEQ city-region have relatively low per capita water demand (compared to the other city-regions), but considerable scope for improving the internalisation of water supply to reduce reliance on 'external' water. Total stormwater runoff, relative to the pre-urbanised reference state, also appears to be low for SEQ compared with the other city-regions. What is hidden in the SEQ results is the temporal characteristics of runoff, which due to the sub-tropical rainfall patterns would be concentrated in more intense runoff events (discussed further in the limitation section). For urban systems in the MEL city-region, there is scope for reducing demand (if water demand in SEQ is used as a benchmark) and for improving internalisation of supply. A stand-out observation for MEL is the high stormwater runoff relative to the nominated pre-urbanised state, suggesting that managing the overall volumes of runoff may be a priority for MEL. Urban systems within the PER city-region have a higher internalisation of water supply than the other city-regions, but reducing water demand would appear to be a priority.

These water performance observations are already understood by water managers in these regions. What the evaluation adds is the ability to quantify performance at the city-region scale to align with urban planning and regional water management processes. The quantified water metabolism performance indicators can be used to make consistent comparisons between city-regions for benchmarking, track changes over time, and identify broad opportunities for improvement. Furthermore they can promote dialogue across stakeholder groups about what the desired performance for the future should look like.

485 The value of presenting the hydrological performance indicators as relative values
486 (normalised against the pre-urbanised reference state) rather than as absolute values warrants
487 discussion. This indicator is a 'distance to target' type of indicator, which judges performance
488 in terms of how far the hydrological flows are from a regionally-relevant reference point
489 rather than the actual scale of the flows. This allows for region-specific reference state to be
490 taken into consideration when judging and comparing performance. While it is recognised
491 that restoring natural hydrological flow to the pre-urbanised state may not be the end
492 objective, it does provide a useful point of reference that takes account regional
493 characteristics. For comparing the urban water performance of the city-regions and the water
494 sensitive interventions, the relative values enabled a more valid comparison would be possible
495 if the results were presented as absolute flows.

496 Evaluating the water sensitive interventions at the city-region scale enabled the
497 relative significance of the individual or combined interventions to be understood. For
498 example, the water sensitive interventions considered here may be considered to be
499 potentially significant, but when considered in the context of the city-region, the water
500 metabolism performance benefits may actually be quite modest. Increasing perviousness to
501 the scale assessed here appears to reduce runoff at the city-region scale only marginally in
502 SEQ and MEL (Figure 4b,d). We observed that combined implementation of a number of
503 interventions would be necessary to make substantive gains in the overall water metabolism
504 performance. Individual initiatives may not have much influence in isolation.

505 A novel contribution of this work was the development of methods for defining the
506 urban system boundary and for estimating hydrological flows at the city-region scale using
507 land use data. This enabled the water flow data for the urban water mass balance to generated
508 in systematic, consistent and repeatable way, so that valid comparisons can be made between

509 different urban systems. Previous water mass balance studies by Kenway et al (2011) of the
510 same regions in 2004/2005 generated urban water flow estimates that were quite different to
511 those estimated in this study for 2013/14. This can be mostly attributed to the fact that the
512 urban system boundaries were different (system boundary areas were 1,281 versus 4,118 km^2
513 for SEQ, 1,818 versus 2,879 km^2 for MEL and 1,200 versus 955 km^2 for PER). The system
514 boundaries of the Kenway et al (2001) study were defined by the available data in each
515 region, and may not have represented comparable systems. The more systematic methods
516 used in this work means that the compared systems can be comparable in terms of the urban
517 entity they represent and the flow data generated to align with this.

518 The water metabolism performance indicators generated at the city-region scale could
519 also be generated alongside other urban water performance indicators (Behzadian and
520 Kapelan 2015a, CRC WSC 2016, van Leeuwen et al. 2012, Venkatesh et al. 2014). In
521 particular, the Water Sensitive Cities (WSC) Index (CRC WSC 2016) provides a framework
522 for reporting urban water performance across biophysical, social and governance aspects. The
523 water metabolism performance indicators demonstrated here could fill a current gap in the
524 quantification of resource management performance objectives in the WSC Index. Reporting
525 the water metabolism indicators within a broader framework also allows barriers and risk
526 factors to be considered, such as public acceptance (Campisano et al. 2017, West et al. 2016)
527 when assessing water sensitive interventions.

528 **4.3 Limitations and future work**

529 One limitation of this work was the estimation of flows as annual averages, which
530 compressed the temporal variations of water flows (eg daily, monthly variation) into a single
531 annual total. This means that the temporal characteristics of runoff is hidden. Consequently,
532 the picture the work provides is a ‘snapshot’ for a given year and for the urban system at that

point in time. To provide a better evaluation and screening of water sensitive interventions, including consideration of storage of water within cities, drawing on data from multiple years, and/or analysis of a finer time-scale (for eg. monthly) is recommended in order to capture the temporal effects.

Similar to the issue of temporal variability of rainfall, spatial variability in this work was limited to an average value of rainfall across the assessed urban systems. Variations in rainfall across this scale could be substantial (perhaps in the order of 200mm average rainfall across the city-region). Future work could build on the method developed here, by partitioning urban systems into a series of representative zones (perhaps based on rainfall districts or some other boundary with meaning to the analysis such as local government areas or catchments).

Another limitation is related to the hydrological flow estimation. The use of land use types and their respective hydrological flow partitioning (HFP) factors was a novel feature of the framework for estimating average hydrological flows at this large scale. They were sufficient for testing the functionality of the framework, but whether these coarse averages are accurate enough for the purpose of screening and prioritisation of interventions still needs to be validated. Further evaluation of the feasibility and preferred location for promising interventions would require detailed hydrological modelling techniques. In future work, we expect the need to estimate hydrological flows for sub-regions using hydrological models.

In this work, we modified a limited selection of variables (water demand, water sources, and perviousness) related to water management interventions to prove the concept. Other variables related to planning interventions could be evaluated using the framework. For instance, estimating the influence of population growth, housing densities, land use

556 distributions, and green spaces, on water metabolism performance would help evaluate
557 planning alternatives in the context of water resources.
558 Climate change is also an important variable to consider in the future applications of
559 the framework. It can lead to temporal redistribution of precipitation (Franczyk and Chang
560 2009), and change in rainfall intensities and runoff peak flow (Zahmatkesh et al. 2015), which
561 influence the benefits of water sensitive interventions. However as noted earlier, temporal and
562 spatial variabilities in water flows are not observable with the framework, and so it may not
563 be the best method for considering climate change influences.

564 The capacity of the developed metabolism framework for informing urban water
565 management could be expanded by considering the other water-related resource efficiency
566 aspects of energy and nutrients, and functionality of water in the urban landscape (Renouf et
567 al. 2017). Only efficiency and hydrological performance was considered here as a first step to
568 understanding water metabolism performance at this scale. In the case of PER, water-related
569 energy is an important consideration given the high use of energy-intensive desalinated water
570 (Figure 4e), and its consideration would amplify the benefits of initiatives that displace
571 desalinated water. For SEQ and MEL, the benefits of internal harvesting of rainwater /
572 stormwater appear similar to those from wastewater recycling (Figure 4 a,c). However, if the
573 nutrient recovery benefits of wastewater recycling were considered (if nutrients are
574 beneficially utilised), wastewater may be more beneficial. The mass balance framework
575 provides a good basis on which to add a water-related energy or water-related nutrient
576 analysis due to its systematic characterisation of all flows.

577 In summary, research needed to further develop the approach would include i) being
578 able to capture the temporal and spatial variation in water flows, ii) validating the HFP factors
579 used to estimate hydrological flows at the city-region scale, iii) exploring a suitable reference

580 case against which to normalise hydrological performance, iv) identifying ways to define
581 sustainable rates of water extractions against which to evaluate the sustainability of 'external'
582 water extraction, v) adding evaluation of water-related energy and nutrient flows, and vi)
583 enabling assessment of climate change implications.

584 **5. CONCLUSIONS**

585 We took a big-picture perspective to the evaluation of the urban water performance,
586 by extending an urban water metabolism evaluation framework to assess urban systems
587 within three Australia city-regions. To test how it can inform water management decisions,
588 we evaluated these regions under current conditions and after the implementation of example
589 water sensitive interventions. In so-doing water performance was evaluated in terms of urban
590 demand for water extracted from the environment and the "hydrological performance" of the
591 urban system.

592 Adapting the evaluation framework to a large urban scale required the development of
593 new methods for i) defining the system boundaries using land use data; and ii) for estimating
594 natural hydrological flows by combining the land use data with hydrological flow partitioning
595 (HFP) factors. The method can be conducted relatively rapidly, and could be used by water
596 managers and urban planners for strategic evaluation. It is transferrable to other city-regions if
597 spatial land use data were available, along with data for water supply and wastewater
598 generation that can be aligned to the defined urban system boundaries.

599 We found that application of the urban water metabolism evaluation framework to
600 urban systems within city-regions is suited to initial screening of the type and scale of water
601 sensitive interventions that may be needed to achieve desired water performance objectives.
602 The work also highlighted the current absence of established performance objectives and

603 prompts the need for urban planners and water managers to consider targets for improvements
604 that are sought at the city-region scale. The values of the adapted approach have been the
605 broader urban perspective (to align with and inform strategic urban and water planning
606 processes), and the systematic definition of the urban system boundary (to enable consistent
607 comparisons between urban systems and over time). The framework can enable a better
608 understanding of the role and impact of water sensitive interventions in achieving water
609 performance objectives for urban systems within city-regions.

610 **Acknowledgement**

611 This work was supported by the Cooperative Research Centre for Water Sensitive
612 Cities (CRC WSC Project B1.2 – "Catchment-scale landscape planning for water sensitive
613 cities in an age of climate change"), which is funded by the Australian Government and
614 industry partners. Marguerite Renouf, Silvia Serrao-Neumann and Edward Morgan were
615 funded by the CRC WSC. Steven Kenway was funded by the Australian Research Council
616 (ARC) through a Discovery Early Career Researcher Award (DECRA DE100101322). The
617 authors acknowledge the input from Jessica Yeung and Mekdes Yimam from the Australian
618 Bureau of Meteorology (BOM) who assisted with the interrogation of the BOM's urban water
619 accounts database; Claire Sotiriadis who compiled the data for the analysis, and Cath Thrupp
620 and Fiona Chandler who reviewed early versions. The very constructive feedback received
621 from the anonymous reviewers was also greatly appreciated.
622

623 **REFERENCES**

624 ABARE (2013) Australian Land Use and Management Classification (Version 7) Australian
625 Bureau of Agricultural and Resource Economics, Canberra.
626 ABARE (2015) Catchment Scale Land Use of Australia - Update May 2016, Australian
627 Bureau of Agricultural and Resource Economics, Canberra.
628 ABS (2013) Environmental Issues: Water use and Conservation, Mar 2013. - Sources of
629 Water and its Uses. ABS Report 4602.0.55.003, Australian Bureau of Statistics, Canberra.
630 ABS (2015) Regional Population Growth, Australia, 2013-14. Catalogue number 3218.0,
631 Australian Bureau of Statistics, Canberra.
632 ADB (2016) Asian Water Development Outlook 2016. Strengthening Water Security in Asia
633 and the Pacific, Asian Development Bank, Asia-Pacific Water Forum, Philippines.
634 Barron, O.V., Barr, A.D. and Donn, M.J. (2013) Effect of urbanisation on the water balance
635 of a catchment with shallow groundwater. Journal of Hydrology 485(SI), 162-176.
636 Behzadian, K. and Kapelan, Z. (2015a) Advantages of integrated and sustainability based
637 assessment for metabolism based strategic planning of urban water systems. Science of the
638 Total Environment 527–528, 220-231.
639 Behzadian, K. and Kapelan, Z. (2015b) Modelling metabolism based performance of an urban
640 water system using WaterMet2. Resources, Conservation and Recycling 99, 84-99.
641 Bhaskar, A.S., Beesley, L., Burns, M.J., Fletcher, T.D., Hamel, P., Oldham, C.E. and Roy,
642 A.H. (2016) Will it rise or will it fall? Managing the complex effects of urbanization on base
643 flow. Freshwater Science 35(1), 293-310.
644 Bhaskar, A.S. and Welty, C. (2012) Water balances along an urban-to-rural gradient of
645 Metropolitan Baltimore, 2001-2009. Environmental & Engineering Geoscience 18(1), 37-50.
646 BOM (2014) National Water Account 2014, Australian Government Bureau of Meteorology.
647 BOM (2015) National Water Account, Australian Government Bureau of Meteorology.
648 BOM (2016) Urban Water Database V2.033, Unpublished database of the Bureau of
649 Meteorology's Urban Water Unit.
650 BOM (2017) Australian Landscape Water Balance database.
651 Brown, R.R. and Farrelly, M.A. (2009) Delivering sustainable urban water management: a
652 review of the hurdles we face. Water Science and Technology 59(5), 839-846.
653 Campisano, A., Butler, D., Ward, S., Burns, M.J., Friedler, E., DeBusk, K., Fisher-Jeffes,
654 L.N., Ghisi, E., Rahman, A., Furumai, H. and Han, M. (2017) Urban rainwater harvesting
655 systems: Research, implementation and future perspectives. Water Research 115, 195-209.

656 Chesterfield, C., Beck, L., Brown, R.R., Dunn, G., de Haan, F., Lloyd, S., Rogers, B.C.,
657 Urich, C. and Wong, T.H.F. (2016) Beyond benchmarking: A water sensitive cities index,
658 Australian Water Association, Melbourne.
659 Coutts, A.M., Tapper, N.J., Beringer, J., Loughnan, M. and Demuzere, M. (2013) Watering
660 our cities: The capacity for Water Sensitive Urban Design to support urban cooling and
661 improve human thermal comfort in the Australian context. Progress in Physical Geography
662 37(1), 2-28.
663 CRC WSC (2016) Water Sensitive Cities Index, Cooperative Research Centre for Water
664 Sensitive Cities, https://watersensitivecities.org.au/solutions/water-sensitive-cities-
665 index/contacts-benchmark-city/.
666 Farooqui, T.A., Renouf, M.A. and Kenway, S.J. (2016) A metabolism perspective on
667 alternative urban water servicing options using water mass balance. Water Research 106, 415-
668 428.
669 Feng, Y., Burian, S. and Pomeroy, C. (2016) Potential of green infrastructure to restore
670 predevelopment water budget of a semi-arid urban catchment. Journal of Hydrology 542, 744-
671 755.
672 Franczyk, J. and Chang, H. (2009) The effects of climate change and urbanization on the
673 runoff of the Rock Creek basin in the Portland metropolitan area, Oregon, USA. Hydrological
674 Processes 23(6), 805-815.
675 Gooda, M.C. and Voogt, S. (2012) Climate Change Impact on Water Availability in the South
676 East Queensland Region., Urban Water Security Research Alliance
677 Government of Western Australia (n.d.) Domestic garden bore FAQs.
678 IWA (2016) The IWA Principles for Water Wise Cities, International Water Association,
679 London.
680 Kennedy, C., Cuddihy, J. and Engel-Yan, J. (2007) The changing metabolism of cities.
681 Journal of Industrial Ecology 11(2), 43-59.
682 Kennedy, C.A., Stewart, I., Facchini, A., Cersosimo, I., Mele, R., Chen, B., Uda, M., Kansal,
683 A., Chiu, A., Kim, K.-g., Dubeux, C., Lebre La Rovere, E., Cunha, B., Pincetl, S., Keirstead,
684 J., Barles, S., Pusaka, S., Gunawan, J., Adegbile, M., Nazariha, M., Hoque, S., Marcotullio,
685 P.J., González Otharán, F., Genena, T., Ibrahim, N., Farooqui, R., Cervantes, G. and Sahin,
686 A.D. (2015) Energy and material flows of megacities. Proceedings of the National Academy
687 of Sciences 112(19), 5985-5990.
688 Kenway, S., Gregory, A. and McMahon, J. (2011) Urban water mass balance analysis.
689 Journal of Industrial Ecology 15(5), 693-706.

690 Khastagir, A. and Jayasuriya, L.N.N. (2010) Impacts of using rainwater tanks on stormwater
691 harvesting and runoff quality. Water Science and Technology 62(2), 324-329.
692 Low Choy, D. and Minnery, J. (1994) Planning the Outward Growth of Australian Capital
693 Cities: Brisbane. Urban Policy and Research 12(3), 200-212.
694 Maheshwari, B., Singh, V. P. and Thoradeniya, B. (eds) (2016) Balanced Urban
695 Development: Options and Strategies for Liveable Cities, Springer.
696 Makropoulos, C.K., Natsis, K., Liu, S., Mittas, K. and Butler, D. (2008) Decision support for
697 sustainable option selection in integrated urban water management. Environmental Modelling
698 & Software 23(12), 1448-1460.
699 Marteleira, R., Pinto, G. and Niza, S. (2014) Regional water flows – Assessing opportunities
700 for sustainable management. Resources, Conservation and Recycling 82(0), 63-74.
701 Newman, P.W.G. (1999) Sustainability and cities: extending the metabolism model.
702 Landscape and Urban Planning 44(4), 219-226.
703 OECD (2012) Redefining 'urban': A new way to measure metropolitan areas, OECD
704 Publishing, Paris.
705 OECD (2015) Water and Cities. Ensuring Sustainable Futures, OECD.
706 Plummer, R., de Grosbois, D., de Loë, R. and Velaniškis, J. (2011) Probing the integration of
707 land use and watershed planning in a shifting governance regime. Water Resources Research
708 47(9).
709 Queensland Government (2016) ShapingSEQ. Draft South East Queensland Regional Plan,
710 Department of Intrastructure Local Government and Planning.
711 Rebitzer, G., Ekvall, T., Frischknecht, R., Hunkeler, D., Norris, G., Rydberg, T., Schmidt,
712 W.P., Suh, S., Weidema, B.P. and Pennington, D.W. (2004) Life cycle assessment: Part 1:
713 Framework, goal and scope definition, inventory analysis, and applications. Environment
714 International 30(5), 701-720.
715 Renouf, M.A. and Kenway, S.J. (2016) Evaluation approaches for advancing urban water
716 goals. Journal of Industrial Ecology 21(4), 995-1009.
717 Renouf, M.A., Serrao-Neuman, S., Kenway, S.J., Low Choy, D. and Morgan, E.A. (2017)
718 Urban water metabolism indicators derived from water mass balance - bridging the gap
719 between vision and performance assessment of urban water resource management. Water
720 Research 122, 669-677.
721 Risbey, J.S. (2011) Dangerous climate change and water resources in Australia. Regional
722 Environmental Change 11(1), 197-203.

723 Rozos, E. and Makropoulos, C. (2013) Source to tap urban water cycle modelling.
724 Environmental Modelling & Software 41(0), 139-150.
725 Schandl, H., Poldy, F., Turner, G.M., Measham, T.G., Walker, D.H. and Eisenmenger, N.
726 (2008) Australia's Resource Use Trajectories. Journal of Industrial Ecology 12(5-6), 669-685.
727 Serrao-Neumann, S., Renouf, M., Kenway, S.J. and Low Choy, D. (2017) Connecting land-
728 use and water planning: Prospects for an urban water metabolism approach. Cities 60, Part B,
729 13-27.
730 Sharma, A.K., Pezzaniti, D., Myers, B., Cook, S., Tjandraatmadja, G., Chacko, P., Chavoshi,
731 S., Kemp, D., Leonard, R., Koth, B. and Walton, A. (2016) Water sensitive urban design: An
732 investigation of current systems, implementation drivers, community perceptions and
733 potential to supplement urban water services. Water 8(7).
734 Sharma, A.K., Tjandraatmadja, G., Cook, S. and Gardner, T. (2013) Decentralised systems -
735 definition and drivers in the current context. Water Science and Technology 67(9), 2091-
736 2101.
737 Thériault, J. and Laroche, A.-M. (2009) Evaluation of the urban hydrologic metabolism of the
738 Greater Moncton region, New Brunswick. Canadian Water Resources Journal / Revue
739 canadienne des ressources hydriques 34(3), 255-268.
740 UK Water Partnership (2015) Future Visions for Water and Cities, A Thought Piece, UK
741 Water Partnership.
742 Umapathi, S., Chong, M.N. and Sharma, A. (2012) Investigation and Monitoring of Twenty
743 Homes to Understand Mains Water Savings from Mandated Rainwater Tanks in South East
744 Queensland, Urban Water Security Research Alliance.
745 Urich, C., Bach, P.M., Sitzenfrei, R., Kleidorfer, M., McCarthy, D.T., Deletic, A. and Rauch,
746 W. (2013) Modelling cities and water infrastructure dynamics. Proceedings of the ICE -
747 Engineering Sustainability 166(5), 01.
748 USDA Soil Conservation Service (1986) Urban Hydrology for Small Watersheds.
749 van Leeuwen, C.J., Frijns, J., van Wezel, A. and van de Ven, F.H.M. (2012) City Blueprints:
750 24 Indicators to Assess the Sustainability of the Urban Water Cycle. Water Resources
751 Management 26(8), 2177-2197.
752 Venkatesh, G., Saegrov, S. and Brattebo, H. (2014) Dynamic metabolism modelling of urban
753 water services - Demonstrating effectiveness as a decision-support tool for Oslo, Norway.
754 Water Research 61(0), 19-33.
755 Victorian Government (2014) Plan Melbourne Metropolitan Planning Strategy, The State of
756 Victoria, Department of Transport Planning and Local Infrastructure, Melbourne.

Vietz, G.J., Rutherfurd, I.D., Fletcher, T.D. and Walsh, C.J. (2016) Thinking outside the channel: Challenges and opportunities for protection and restoration of stream morphology in urbanizing catchments. Landscape and Urban Planning 145(34-44).

Walton, A. and Hume, M. (2011) Creating positive habits in water conservation: the case of the Queensland Water Commission and the Target 140 campaign. Journal of Nonprofit and Voluntary Sector 224 (August), 215–224.

Water Corporation (n.d.) Groundwater replenishment.

West, C., Kenway, S., Hassall, M. and Yuan, Z. (2016) Why do residential recycled water schemes fail? A comprehensive review of risk factors and impact on objectives. Water Research 102, 271-281.

Western Australian Government (2015) Perth-Peel@3million, Western Australian Planning Commission, Perth.

Whiteoak, K., Jones, P. and Pickering, P. (2012) Progress against the national target of 30% of Australia's wastewater being recycled by 2015, Marsden Jacobs Associates, Perth.

Wolman, A. (1965) The Metabolism of Cities. Scientific American 213(3), 179-190.

Wong, T.H.F. and Brown, R.R. (2009) The water sensitive city: principles for practice. Water Science and Technology 60(3), 673-682.

Zahmatkesh, Z., Burian, S.J., Karamouz, M., Tavakol-Davani, H. and Goharian, E. (2015) Low-impact development practices to mitigate climate change effects on urban stormwater runoff: Case study of New York City. Journal of Irrigation and Drainage Engineering 141(1).

Figure captions

Figure 1: Urban water metabolism evaluation framework

Based on an urban water mass balance approach originally described by Kenway et al. (2011) and further developed by Farooqui (2016).

Figure 2: Location of the case study Australian city-regions and the defined urban system areas within them

The red boundary lines denote the extent of the 'city-regions' as described in strategic regional plans. Orange shading shows the area of 'urban systems' within the city-regions, comprising urban and peri-urban land uses (as at 2014).

Figure 3 Urban water metabolism indicators

(WS 1: reduced demand; WS 2: internal harvesting; WS 3: wastewater recycling; WS 4: increase perviousness)

796 **Table 1 Features of the case study regions**

	South East Queensland (SEQ)	Melbourne (MEL)	Perth (PER)
Area of urban systems within the city-region (km^2)	4,318	2,879	955
Climate	Sub-tropical / wet Heavy summer rain	Temperate / wet Rainfall distributed over the year	Temperate / very dry Very dry summer
Rainfall (2013/14) [1]	681mm	845mm	771mm
Major water sources (2013/14)	Surface water	Surface water	Groundwater Surface water, Seawater desalination

797 Notes:
798 1. From Australian Bureau of Meteorology (BOM 2014)
799

800

801 **Table 2 Summary of data sources**

Data	Source	Year of data
Land use data for defining the spatial extent of urban systems within the city-regions	Catchment Scale Land Use of Australia (CLUM) database Version 7 (ABARE 2015), using the Australian Land Use and Management Classifications (ALUM) system (ABARE 2013)	2014
Managed water flows (centralised)		
- Centralised potable water (surface water, groundwater, desalinated)	Australian Bureau of Meteorology's (BOM) Urban Water Accounts database (BOM 2016) - data for the water utility jurisdictions present within the defined urban system area were extracted from the database	2013-2014
- Wastewater discharged		
- Wastewater recycled		
Managed water flows (decentralised)		
- Harvested rainwater	Published survey of the extent of rainwater harvesting in the city-regions (ABS 2013), from which average rates of harvesting for the defined urban system area were derived.	2013
- Harvested runoff	Published estimates (Whiteoak et al. 2012)	2012
- Private bore water	Published estimates (Government of Western Australia n.d.)	
Natural water hydrological flows		
- Precipitation	Total area-averaged annual rainfall database (BOM 2014) for the defined urban system area. It is compiled by BOM from weather stations within the respective regions.	2013-2014
- Runoff coefficients	Derived from the U.S. Soil Conservation Service runoff curve number method (USDA Soil Conservation Service 1986)	2013-2014
- Infiltration coefficients	Derived from the annual deep drainage reported in the BOM Australian Landscape Water Balance (BOM 2017)	2013-2014
- Evapotranspiration coefficients	Remainder after deducting runoff and infiltration	2013-2014
Urban population	Australian Bureau of Statistics (ABS 2015) – populations for the statistical local areas within the urban system area.	2013-2014

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804 **Table 3 Hydrological flow partitioning (HFP) factors**

Land use category	SEQ			MEL			PER		
	ET	G	SW	ET	G	SW	ET	G	SW
Urban residential	0.66	0.03	0.31	0.86	0.05	0.09	0.78	0.17	0.05
Manufacturing	0.64	0.03	0.33	0.84	0.05	0.11	0.76	0.16	0.08
Commercial and public services	0.55	0.02	0.42	0.76	0.04	0.19	0.66	0.14	0.19
Roads, airports, transport and communications	0.34	0.01	0.65	0.50	0.03	0.47	0.30	0.07	0.63
Utilities and municipal services	0.73	0.03	0.24	0.90	0.05	0.04	0.81	0.18	0.02
Mining, extractive industries	0.73	0.03	0.24	0.90	0.05	0.04	0.81	0.18	0.02
Recreation and cultural	0.85	0.04	0.12	0.93	0.05	0.02	0.81	0.18	0.01
Natural environment	0.87	0.04	0.09	0.93	0.05	0.02	0.81	0.18	0.01
Rural residential	0.83	0.04	0.14	0.93	0.05	0.02	0.81	0.18	0.01

805 *Totals may not add up to one due to rounding. Details of estimating these HFP factors can be found in the Supplementary*
806 *Material.*
807
808

809 **Table 4 Urban water metabolism indicators**

Indicator	Equation
Urban water efficiency per capita Total use of 'external' water per capita per year (kL/capita/yr)	$\frac{C}{\text{Population}}$
Water supply internalisation Proportion of total urban water demand met by internally harvested / recycled water)	$\frac{D + Re}{D + Re + C}$
Hydrological performance Ratio of post (i) to pre (o) urbanised annual flows of stormwater runoff, evapotranspiration, and infiltration to groundwater	$\frac{SWi}{SWo}, \frac{Gi}{Go}, \frac{ETi}{ETo}$

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812 **Table 5 Water sensitive interventions**

		Variables				
Region	Water sensitive interventions	Urban water demand (kL/p/yr)	Percentage of households with a rainwater tank installed	Percentage of harvestable stormwater[1] that is harvested	Percentage of wastewater recycled	Hydrological flow partitioning factors of the urban residential areas (ET:G:SW)
SEQ	Pre-urbanised	NA	NA	NA	NA	0.87 : 0.04 : 0.09[2]
	Current (2013/14)	92	47%	0.04%	7%	0.66 : 0.03 : 0.31
	WS-1 Reduced demand	85	-	-	-	-
	WS-2 Internal harvesting	-	60%	5%	-	-
	WS-3 Wastewater recycling	-	-	-	20%	-
	WS-4 Increase perviousness	-	-	-	-	0.77 : 0.03 : 0.20
MEL	Pre-urbanised	NA	NA	NA	NA	0.93 : 0.05 : 0.02[2]
	Current (2013/14)	107	31%	0.30%	10%	0.86 : 0.05 : 0.09
	WS-1 Reduced demand	91	-	-	-	-
	WS-2 Internal harvesting	-	40%	5%	-	-
	WS-3 Wastewater recycling	-	-	-	20%	-
	WS-4 Increase perviousness	-	-	-	-	0.92 : 0.05 : 0.02
PER	Pre-urbanised	NA	NA	NA	NA	0.81 : 0.18 : 0.01[2]
	Current (2013/14)	182	9%	0.00%	6%	0.78 : 0.17 : 0.05
	WS-1 Reduced demand	163	-	-	-	-
	WS-2 Internal harvesting	-	-	-	-	-
	WS-3 Wastewater recycling	-	-	-	40%	-
	WS-4 Increase perviousness	-	-	-	-	0.92 : 0.05 : 0.02

813 Notes:
814 [1] Harvestable stormwater potential is the amount of stormwater runoff that could be harvested without compromising natural
815 hydrological flows, i.e., total estimated runoff (from the urbanised state) minus estimated runoff from the pre-urbanised state.
816 [2] The hydrological partitioning factors (HFP) for the pre-urbanised case were assumed to be that of the"natural
817 environment".
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Table 6 – Urban water mass balance for urban systems within the case study city-regions (flows in GL/yr) (flows that changed relative to the current case are shaded)

	Water demand	Inflows								Outflows				Re	Mass Balance		
	C + D + Re	P	C	(C_s)	(C_g)	(C_d)	D	(D_r)	(D_s)	ET	SW	G	W		Qi (P+C+ D)	Qo (ET + SW + G + W)	ΔS
Urban systems within SEQ city-region																	
Pre-urbanised	-	2,804	-	-	-	-	-	-	-	2,397	298	109	-	-	2,804	2,804	0
Current (2013/14)	304	2,790	261	(260)	(1)	(0.2)	14	(13.9)	(0.2)	2,184	599	100	182	29	3 094	3,094	0
WS1-Reduced demand	283	2,790	240	(239)	(1)	(0.2)	14	(13.9)	(0.2)	2,184	599	100	161	29	3,073	3,073	0
WS2-Internal harvest	304	2,758	229	(227)	(1)	(0.2)	46	(17.7)	(28.5)	2,181	570	100	182	29	3,062	3,062	0
WS3-Recycling	304	2,790	233	(232)	(1)	(0.2)	14	(13.9)	(0.2)	2,184	599	100	154	57	3,094	3,094	0
WS4-Increased perv	304	2,790	261	(260)	(1)	(0.2)	14	(13.9)	(0.2)	2,255	525	104	182	29	3,094	3,094	0
Urban systems within MEL city-region																	
Pre-urbanised	-	2,433	-	-	-	-	-	-	-	2,253	49	130	-	-	2,433	2,433	0
Current (2013/14)	494	2,415	415	(415)	(0)	(0)	18	(16)	(2)	2,153	286	124	285	61	2,908	2,908	0
WS1-Reduced demand	421	2,415	342	(342)	(0)	(0)	18	(16)	(2)	2,153	286	124	212	61	2,836	2,836	0
WS2-Internal harvest	494	2,377	377	(377)	(0)	(0)	55	(20)	(35)	2,149	253	123	285	61	2,871	2,871	0
WS3-Recycling	494	2,415	377	(377)	(0)	(0)	18	(16)	(2)	2,153	286	124	247	99	2,908	2,908	0
WS4-Increased perv	494	2 415	415	(415)	(0)	(0)	18	(16)	(2)	2,205	232	127	285	61	2,908	2,908	0
Urban systems within PER city-region																	
Pre-urbanised	-	742	-	-	-	-	-	-	-	601	10	131	-	-	742	742	0
Current (2013/14)	369	740	268	(30)	(125)	(113)	90	(90)	(0)	766	35	167	131	10	1,109	1,109	0
WS1-Reduced demand	329	740	228	(30)	(125)	(73)	90	(90)	(0)	766	35	167	91	10	1,069	1,069	0
WS2-Internal harvest	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
WS3-Recycling	369	740	240	(30)	(125)	(85)	90	(90)	(0)	766	35	167	103	38	1,109	1,109	0
WS4-Increased perv	369	740	268	(30)	(125)	(113)	90	(90)	(0)	784	13	171	131	10	1,109	1,109	0

Notes:
City-regions: SEQ = South East Queensland; MEL= Melbourne; PER=Perth
Cases: 'Pre-urbanised'= conditions assumed to be in place prior to urbanisation, i.e. a natural landscape; 'Current' = approximation of current conditions based on 2013/14 data; 'WS'= hypothetical water sensitive interventions, (see Table 4).
Acronyms for water flows: P= unharvested precipitation, C = total centralised (external) water supply, C_s = centralised (external) supply from surface waters, C_g = centralised (external) supply from groundwater, C_d = centralised (external) supply of desalinated water, D = total de-centralised (internal) water supply, D_r = de-centralised (internal) supply from harvested rainwater, D_s = de-centralised (internal) supply from harvested stormwater, ET = evapotranspiration, SW = stormwater runoff, G = groundwater infiltration, W= wastewater discharged, Re = recycled wastewater, Qi = total inflows, Qo = total outflows, ΔS = change in storage.

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Figure 1

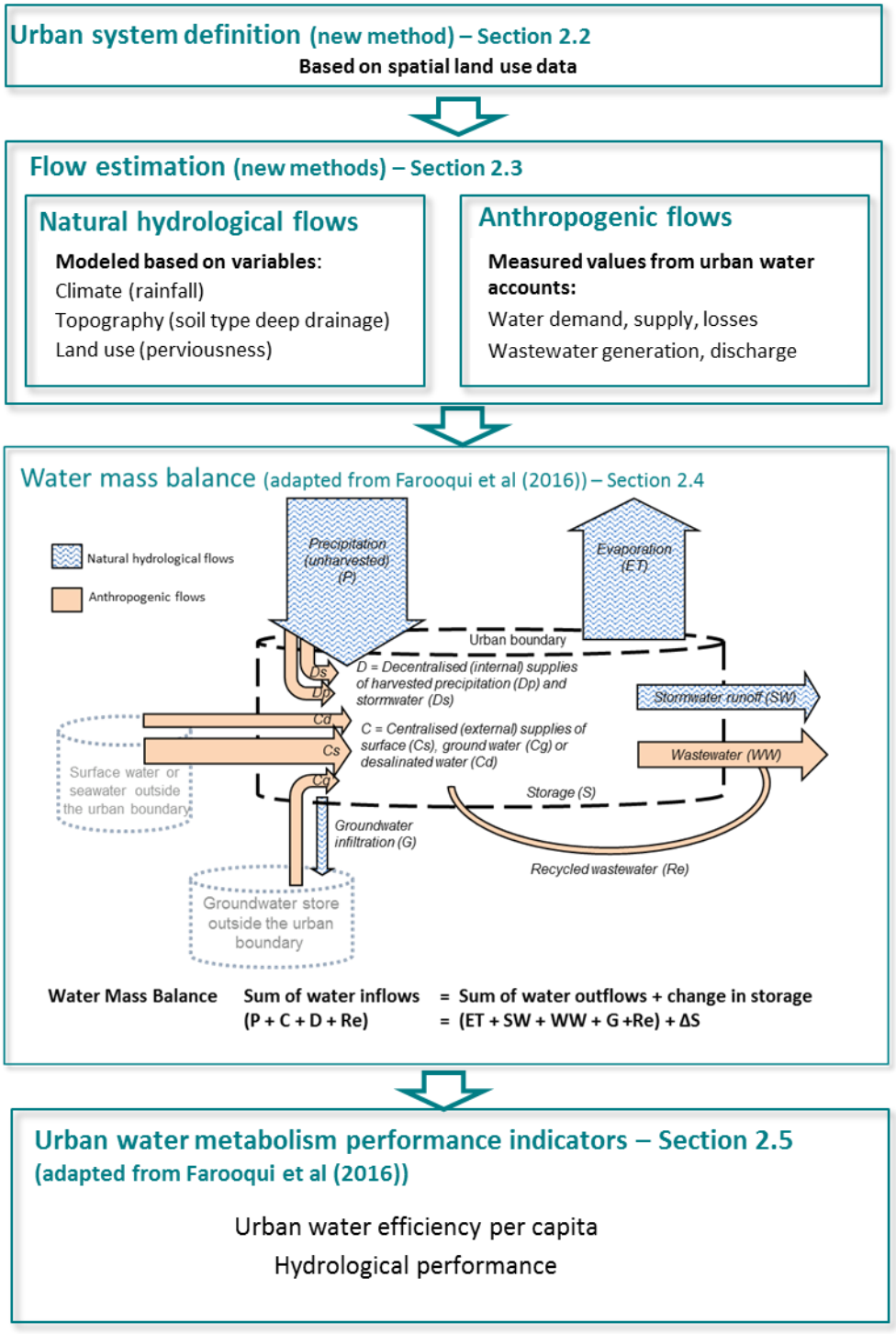

Urban system definition (new method) – Section 2.2
Based on spatial land use data
Flow estimation (new methods) – Section 2.3
Natural hydrological flows
Modeled based on variables:
Climate (rainfall)
Topography (soil type deep drainage)
Land use (perviousness)
Anthropogenic flows
Measured values from urban water accounts:
Water demand, supply, losses
Wastewater generation, discharge
Water mass balance (adapted from Farooqui et al (2016)) – Section 2.4
Natural hydrological flows
Anthropogenic flows
Precipitation (unharvested) (P)
Evaporation (ET)
Urban boundary
Ds
Dp
D = Decentralised (internal) supplies of harvested precipitation (Dp) and stormwater (Ds)
Stormwater runoff (SW)
Cd
Cs
C = Centralised (external) supplies of surface (Cs), ground water (Cg) or desalinated water (Cd)
Wastewater (WW)
Surface water or seawater outside the urban boundary
Cg
Storage (S)
Groundwater infiltration (G)
Recycled wastewater (Re)
Groundwater store outside the urban boundary
Water Mass Balance
Sum of water inflows (P + C + D + Re)
= Sum of water outflows + change in storage
= (ET + SW + WW + G +Re) + ΔS
Urban water metabolism performance indicators – Section 2.5
(adapted from Farooqui et al (2016))
Urban water efficiency per capita
Hydrological performance

Figure 2

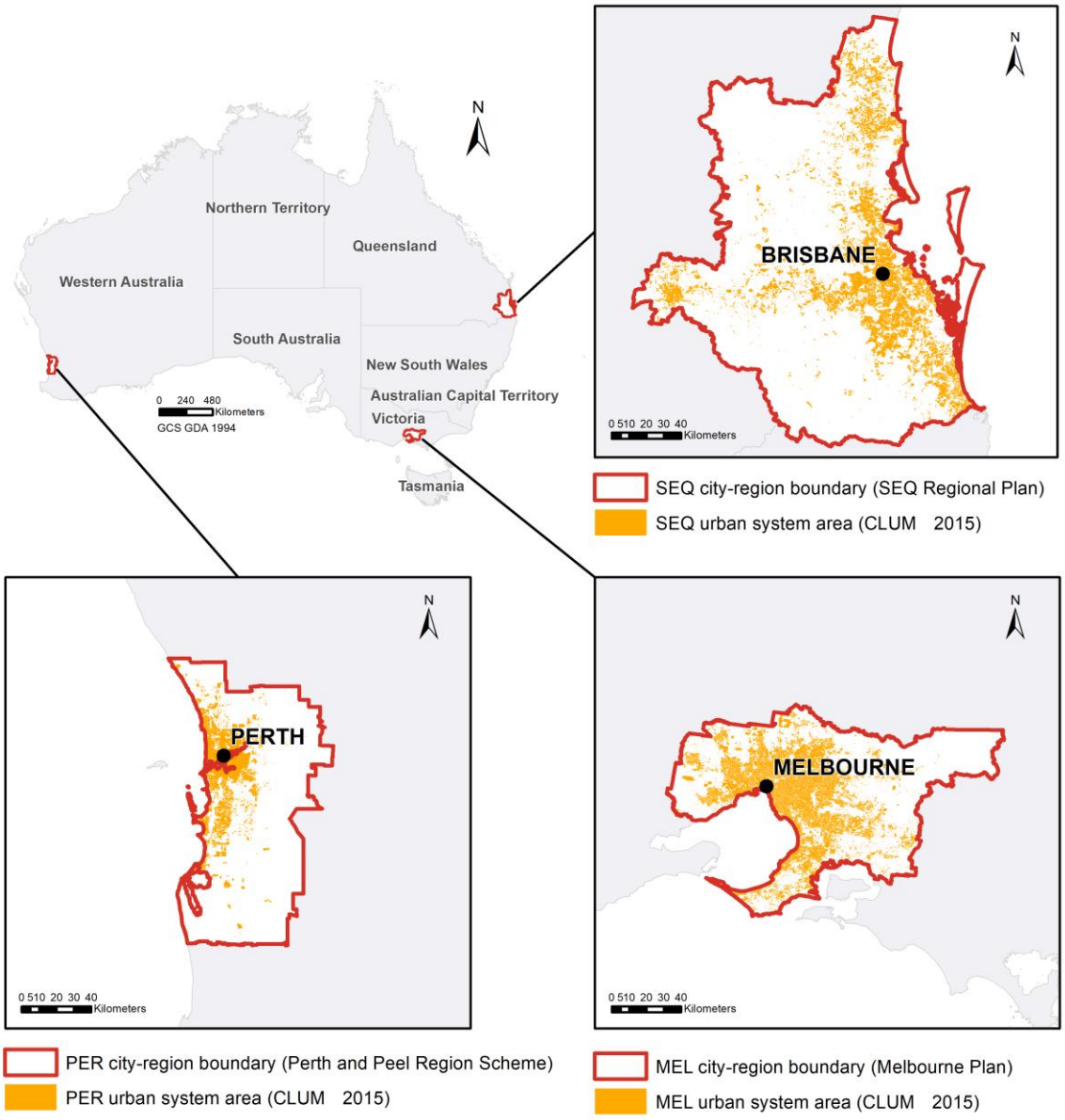
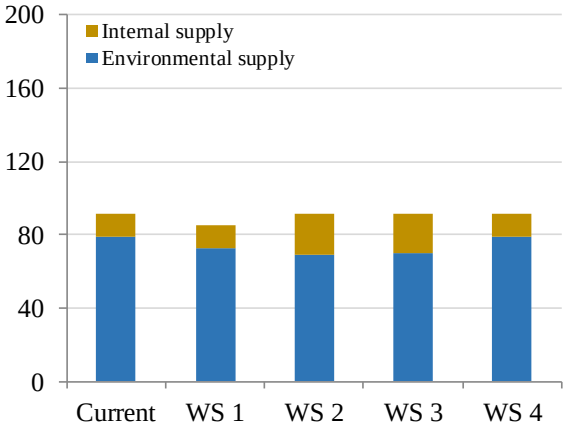
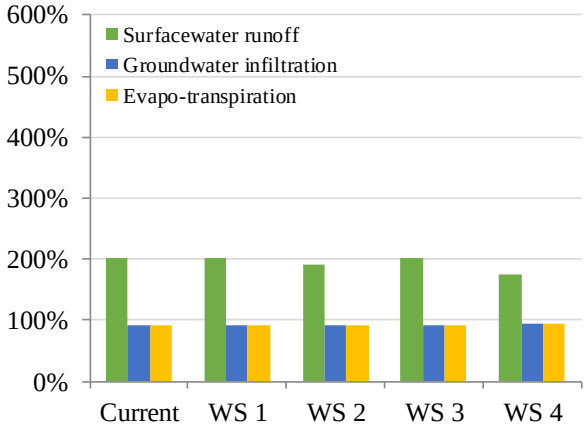

N
Northern Territory
Queensland
Western Australia
South Australia
New South Wales
Australian Capital Territory
Victoria
Tasmania
0 240 480
Kilometers
GCS GDA 1994
BRISBANE
0 510 20 30 40
Kilometers
SEQ city-region boundary (SEQ Regional Plan)
SEQ urban system area (CLUM 2015)
PERTH
PER city-region boundary (Perth and Peel Region Scheme)
PER urban system area (CLUM 2015)
MELBOURNE
MEL city-region boundary (Melbourne Plan)
MEL urban system area (CLUM 2015)

Figure 3

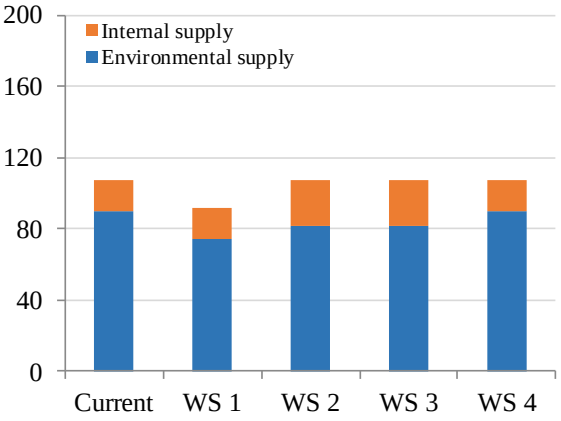
a) **SEQ** urban water efficiency (kL/p/yr), and water supply internalisation



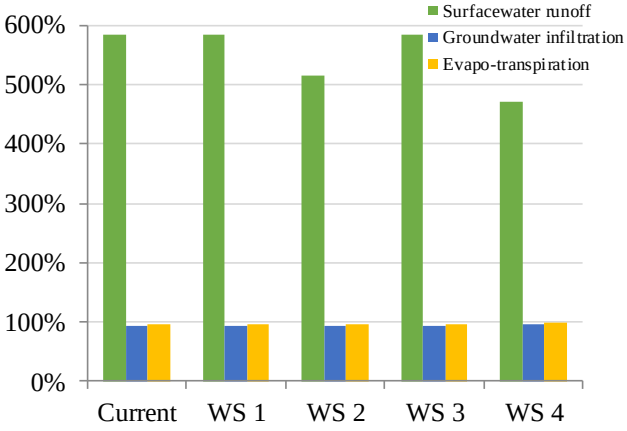
b) **SEQ** hydrological performance (% of pre-urbanised flows)



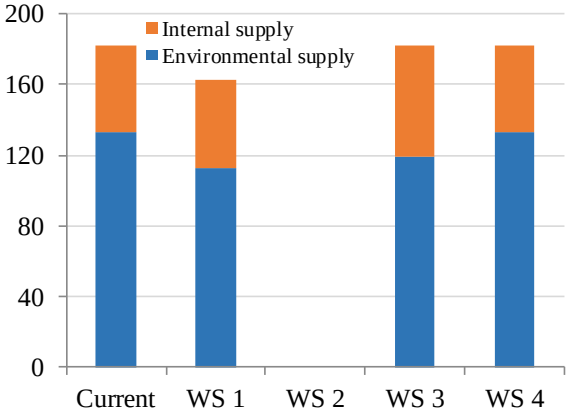
c) **MEL** urban water efficiency (kL/p/yr), and water supply internalisation



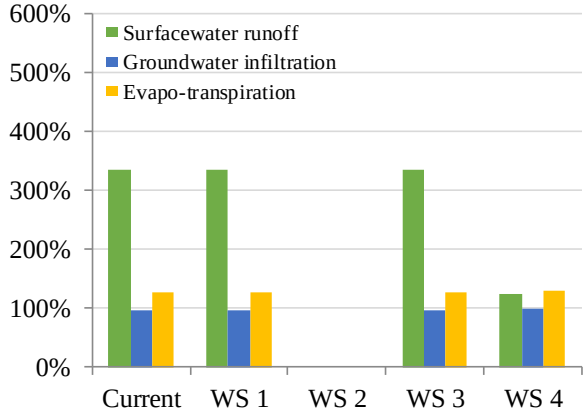
d) **MEL** hydrological performance (% of pre-urbanised flows)



e) **PER** urban water efficiency (kL/p/yr), and water supply internalisation



f) **PER** hydrological performance (% of pre-urbanised flows)



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